

# Balireddy Mahalakshmi

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## PROFESSIONAL SUMMARY

AI/ML Graduate with a strong foundation in building intelligent systems that solve real-world problems. I specialize in developing deep learning models—like YOLO and LSTMs—and connecting them to live data sources. I have hands-on experience in Python automation, predictive analytics, and deploying solutions using AWS cloud. I'm passionate about making AI practical, scalable, and easy to use in everyday life.

## TECHNICAL SKILLS

- **Programming & Databases:** Python, SQL (MySQL), C, Data Structures & Algorithms (DSA).
- **Machine Learning & AI:** Supervised/Unsupervised Learning, Deep Learning (CNN, LSTM), Natural Language Processing (NLP)
- **Frameworks & Libraries:** TensorFlow, Keras, OpenCV, Scikit-learn, Pandas, NumPy, Matplotlib.
- **Software Engineering Tools:** Git, GitHub (Version Control), VS Code, API Development (FASTAPI)
- **Data Visualization & Analysis:** Power BI, MS Excel, Exploratory Data Analysis (EDA).
- **Cloud Computing:** AWS (Certified Cloud Practitioner), Model Deployment Fundamentals.

## PROFESSIONAL EXPERIENCE

**Machine Learning Intern** | FabHost Tech Solutions | Jan 2023 – Mar 2023

- Developed and optimized supervised machine learning models (Random Forest, XGBoost) for predictive analytics using **Python** and **Scikit-learn**.
- Engineered data **preprocessing pipelines** and performed advanced **Feature Engineering**, resulting in a **10% increase in model accuracy** and a significant reduction in bias.
- **Automated data-cleaning workflows** by developing custom scripts and integration pipelines, reducing manual data preparation time by **20%**.
- **Streamlined ML workflows** by integrating modular code practices, ensuring scalability and reproducibility of experimental results across the team.

## PROJECTS

### IoT-Based Weather Forecasting System

- Developed a **Time-Series Forecasting model** using **LSTM (Long Short-Term Memory)** neural networks in **TensorFlow**, achieving high predictive accuracy for short-term weather trends.
- Engineered an **interactive visualization layer** by integrating the **Google Maps API**, enabling location specific data mapping and real-time dashboard updates.
- Implemented **automated event-triggering** via **IFTTT protocols**, providing instant mobile notifications when environmental thresholds were exceeded.
- **Optimized data ingestion pipelines** to handle asynchronous sensor streams, ensuring minimal latency between data collection and model inference.

### Netra: AI-Based Real-Time Assistive Vision System

- Developed a **multi-modal assistive system** for the visually impaired, integrating **Object Detection (YOLOv8)**, **OCR (EasyOCR)**, and **Image Classification** into a unified AI pipeline.
- Architected a **Client-Server AI infrastructure** using **Android (CameraX)** for real-time frame capture and a **FastAPI backend** for high-speed model inference.
- **Asynchronous API calls (Retrofit)**, ensuring seamless audio feedback via Text-to-Speech (TTS).
- Engineered robust data processing workflows using **OpenCV** for image preprocessing (noise reduction, thresholding) and **MobileNetV2** for high-accuracy currency recognition.

## EDUCATION

### **Bachelor of Technology, AI & Machine Learning**

2021-2025

*Saveetha Engineering College, Chennai, TN | CGPA: 8.0/10*

### **Intermediate, MPC**

2019-2021

*Sri Chaitanya Junior College, Andhra Pradesh | Percentage: 80%*

## CERTIFICATIONS

- AWS Certified Cloud Practitioner – Udemy
- Machine Learning with Python – IBM / Udemy
- Python Programming Certification – FabHost Tech Solutions

## ACHIEVEMENTS & LEADERSHIP

- Drowsiness Detection System: Engineered a CNN-based Deep Learning model achieving 95% accuracy in real-time driver fatigue monitoring.
- Internship Optimization: Improved model performance by 10% at FabHost through advanced Feature Engineering and data pipeline refinement.
- Event Leadership: Served as an Organizer for Celenza 2K24, a national-level technical symposium; managed logistics and technical workshops for 500+ participants.
- Hackathon Participant: Competed in AI-focused hackathons, developing rapid prototypes for real-world predictive analytics challenges.